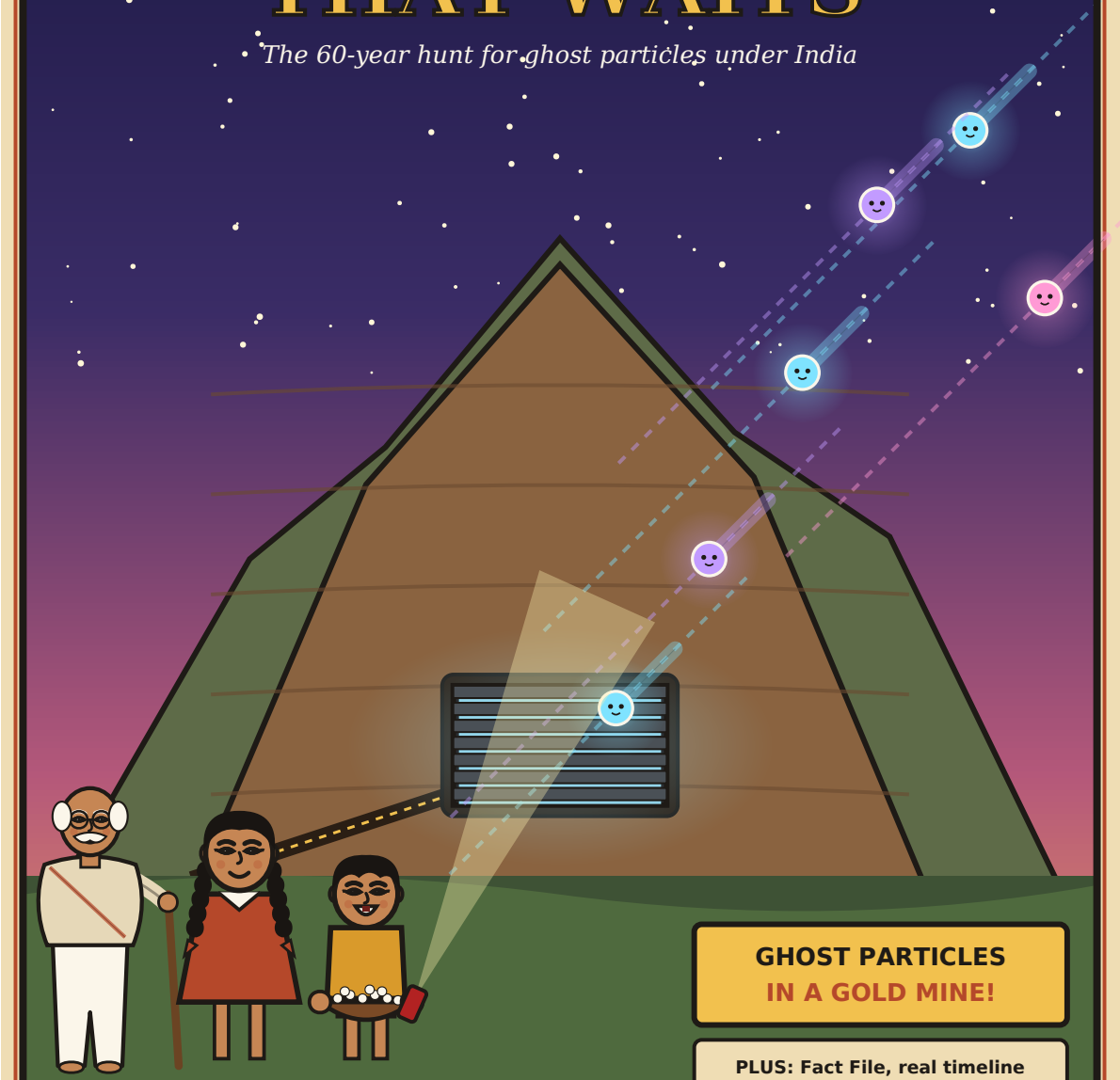
TARAR GOLPO

Science stories from Dadu's verandah

ISSUE 2 OCTOBER 2026

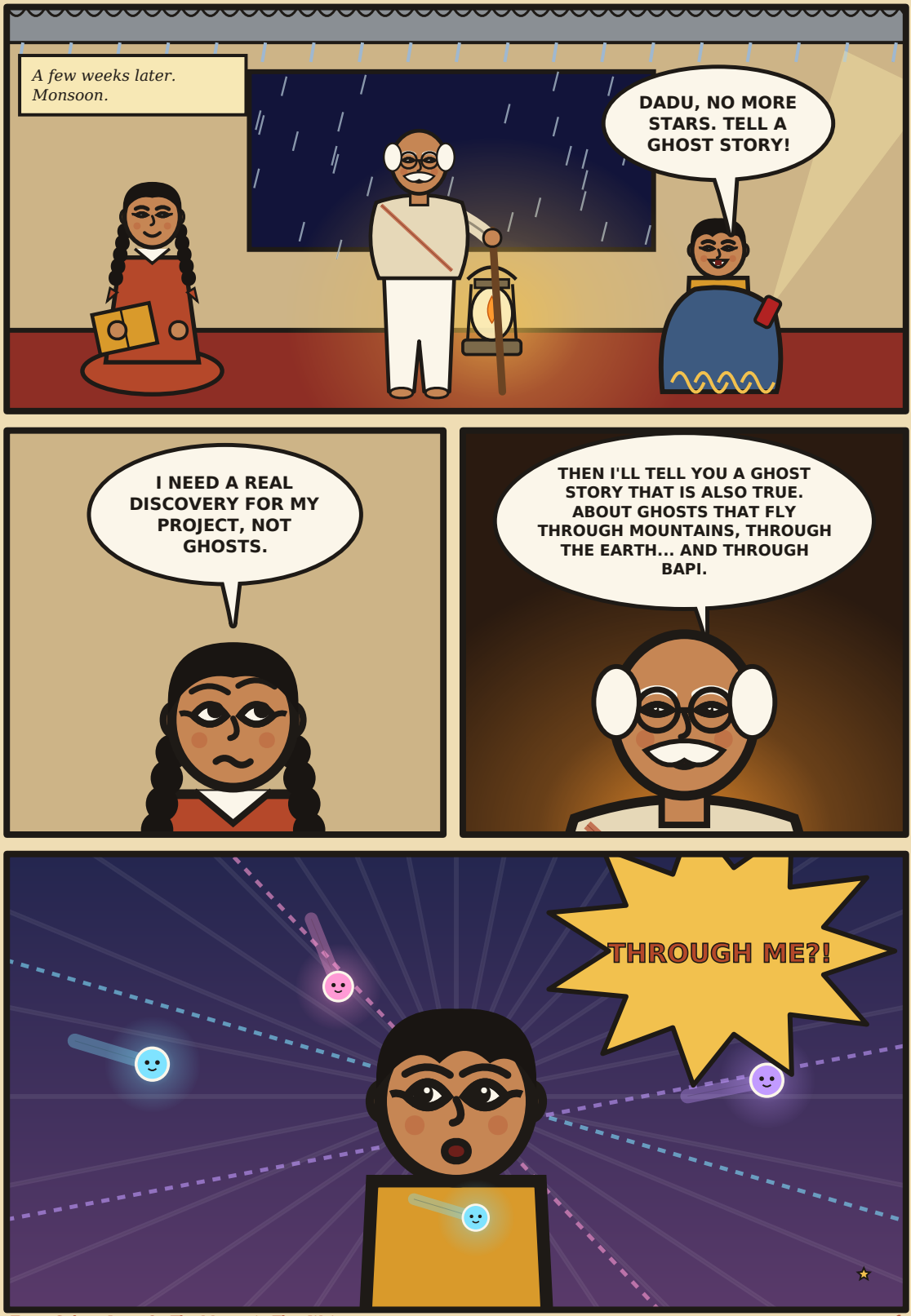
THE MOUNTAIN THAT WAITS

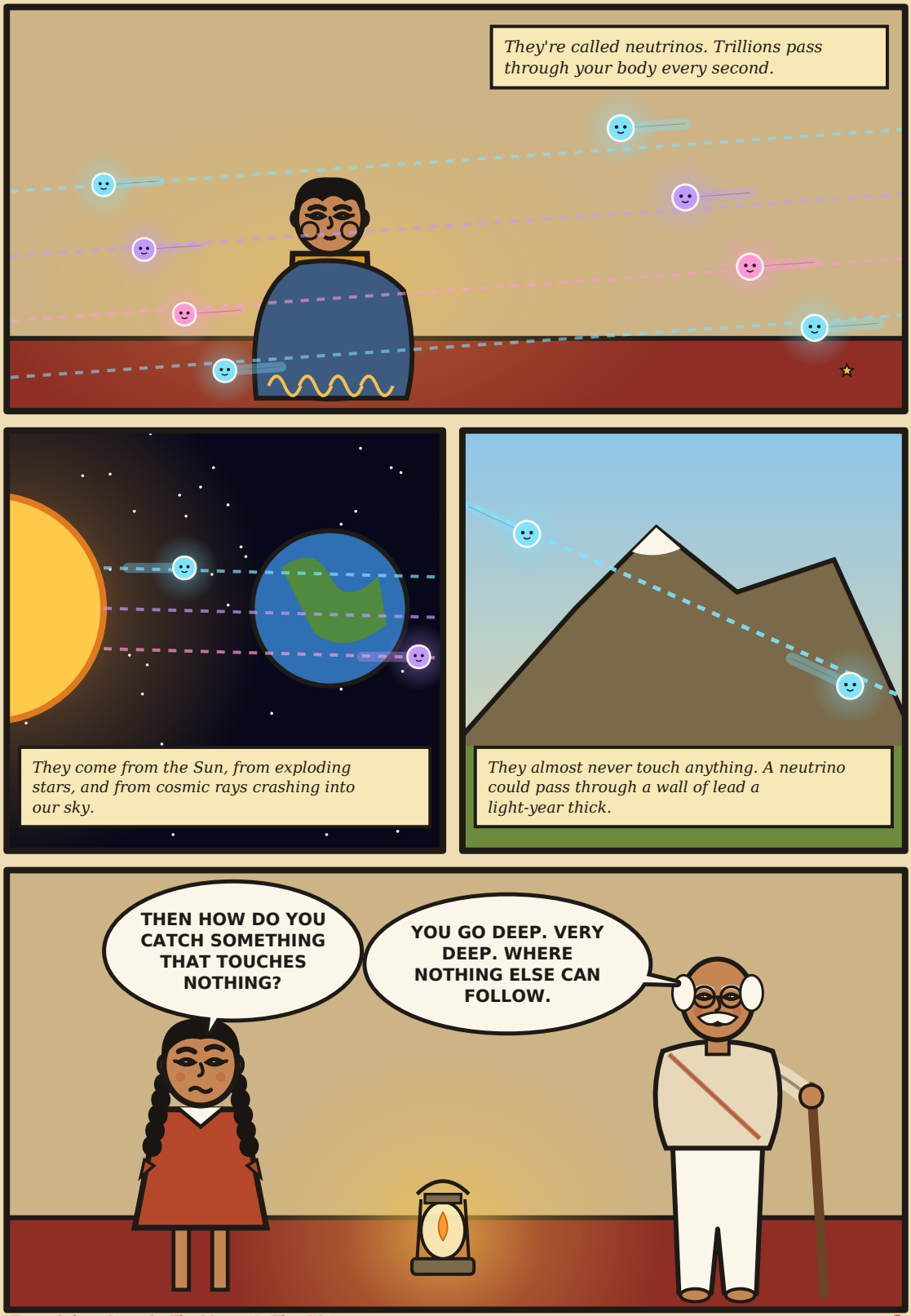
The 60-year hunt for ghost particles under India



GHOST PARTICLES IN A GOLD MINE!

PLUS: Fact File, real timeline and words to know


A few weeks later.
Monsoon.
DADU, NO MORE STARS. TELL A GHOST STORY!
I NEED A REAL DISCOVERY FOR MY PROJECT, NOT GHOSTS.
THEN I'LL TELL YOU A GHOST STORY THAT IS ALSO TRUE. ABOUT GHOSTS THAT FLY THROUGH MOUNTAINS, THROUGH THE EARTH... AND THROUGH BAPI.
THROUGH ME?!


They're called neutrinos. Trillions pass through your body every second.
They come from the Sun, from exploding stars, and from cosmic rays crashing into our sky.
They almost never touch anything. A neutrino could pass through a wall of lead a light-year thick.
THEN HOW DO YOU CATCH SOMETHING THAT TOUCHES NOTHING?
YOU GO DEEP. VERY DEEP. WHERE NOTHING ELSE CAN FOLLOW.

Kolar Gold Fields, Karnataka. One of the deepest mines on Earth.

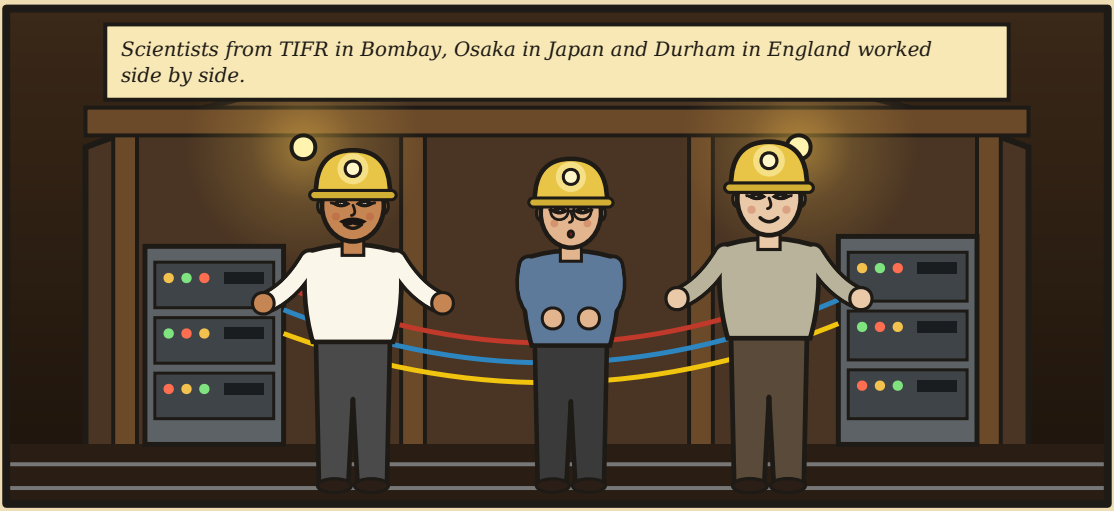
My uncle worked there. One day he told me: "There are scientists down below. Not looking for gold."

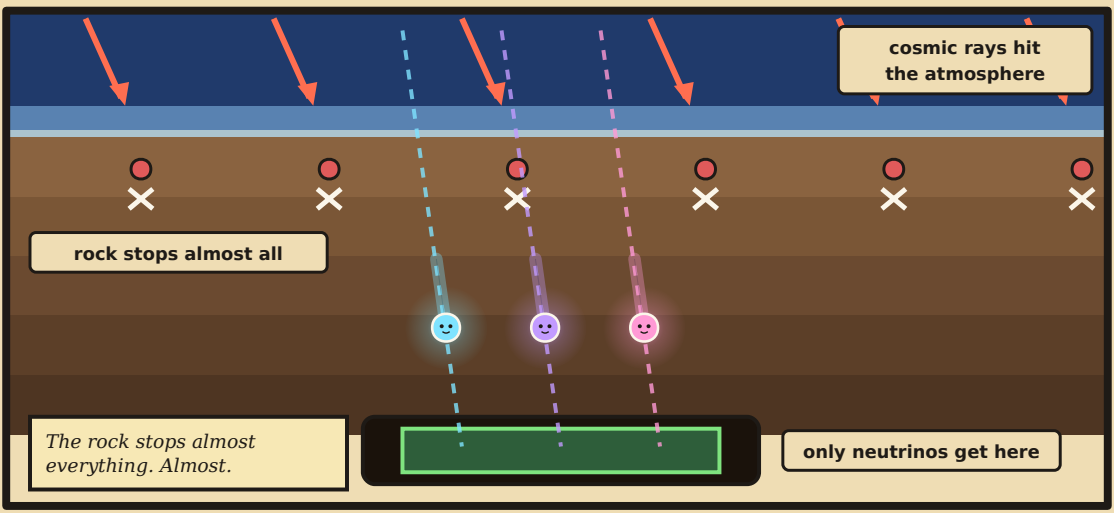
HOLD TIGHT, SCIENTIST-SAHIB! IT'S A LONG WAY DOWN.

About two kilometres down. Two kilometres of rock above their heads.

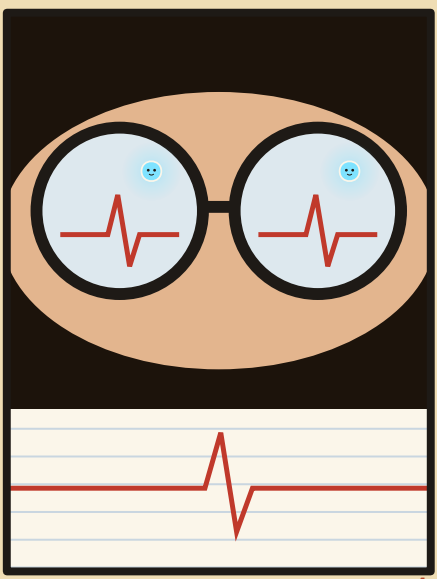
≈ 2 km

Scientists from TIFR in Bombay, Osaka in Japan and Durham in England worked side by side.

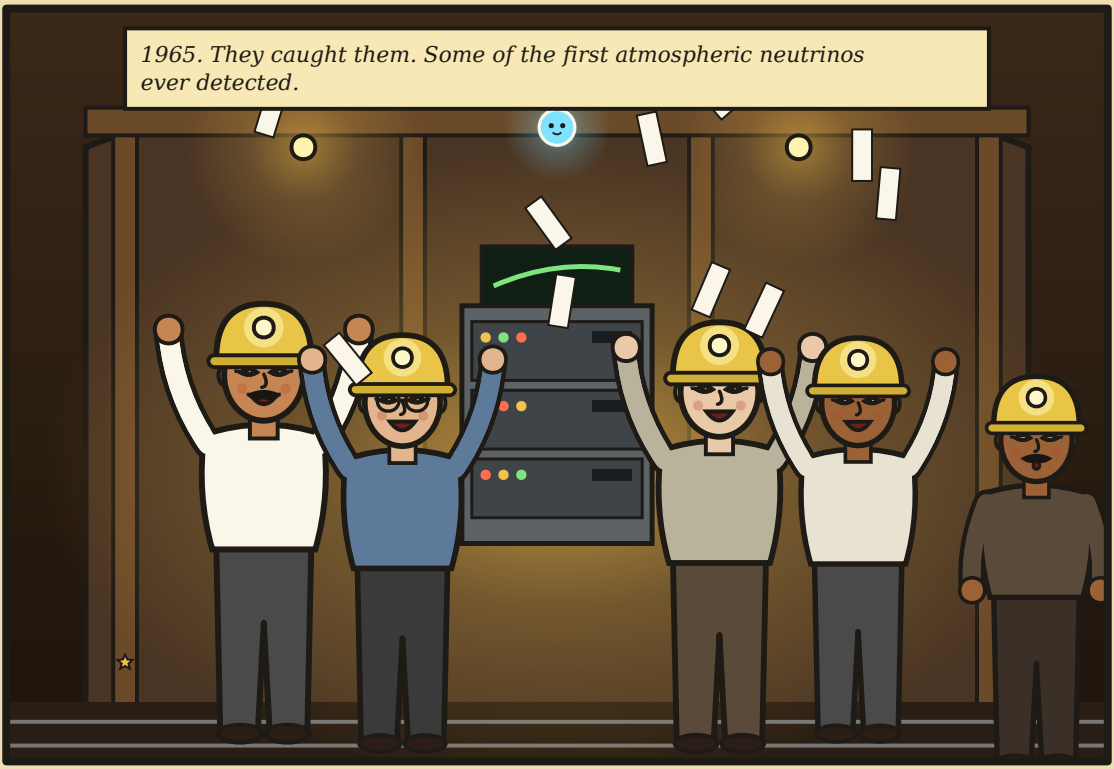




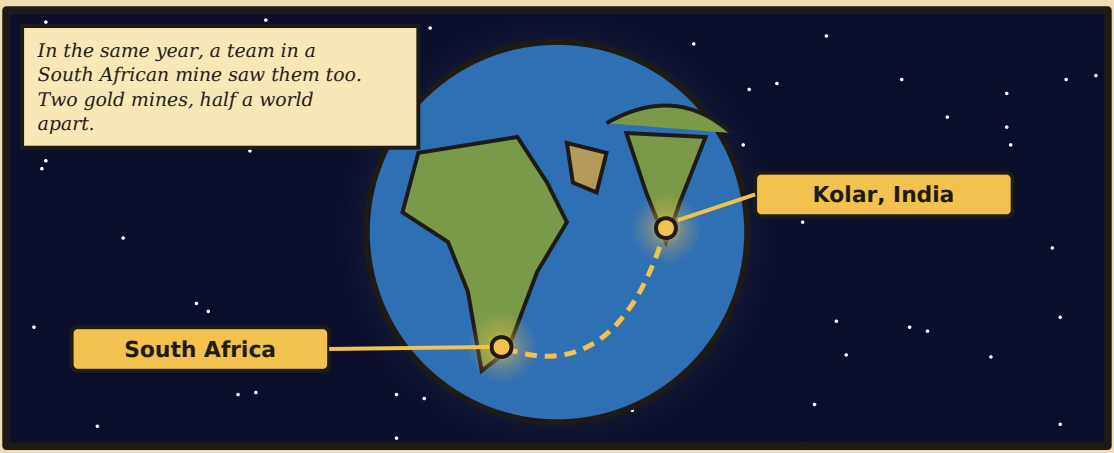




1965. They caught them. Some of the first atmospheric neutrinos ever detected.

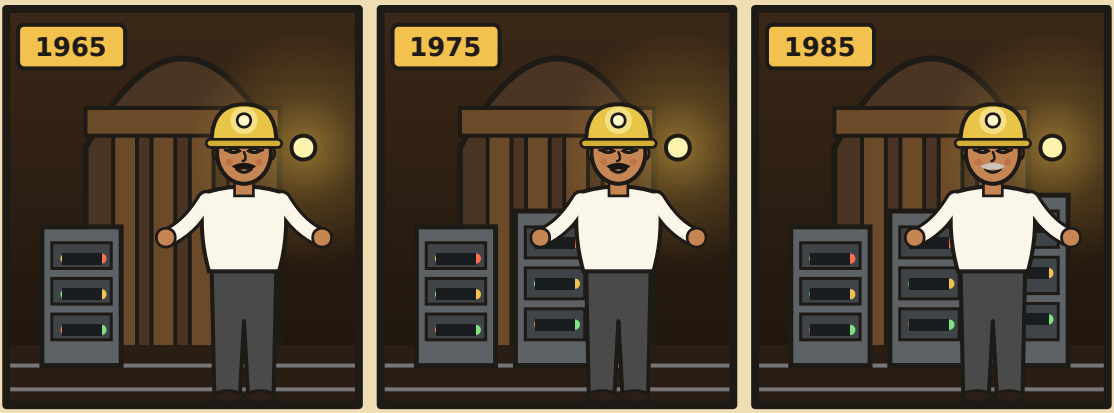


In the same year, a team in a South African mine saw them too. Two gold mines, half a world apart.

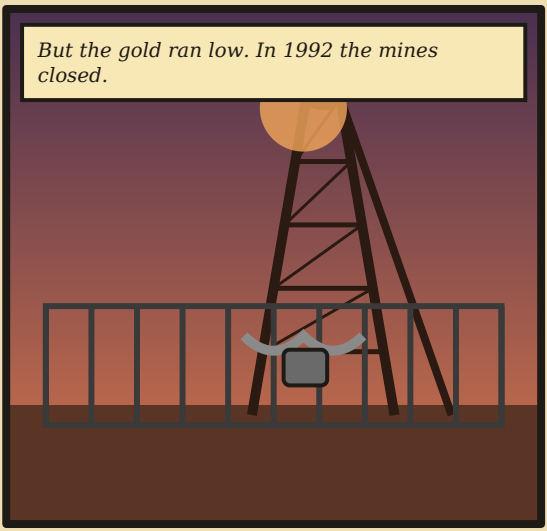




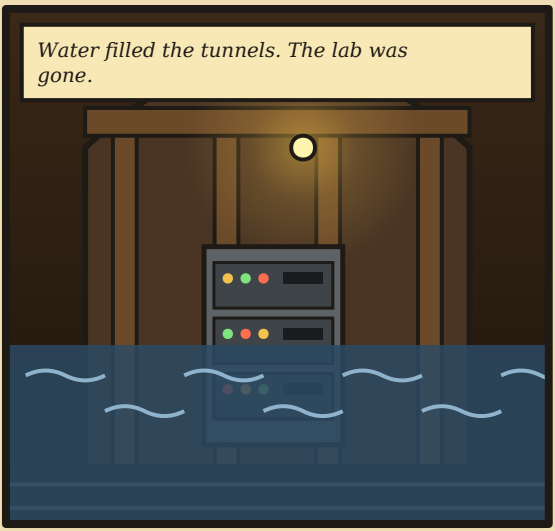
For almost thirty years, Kolar was one of the world's great underground labs.



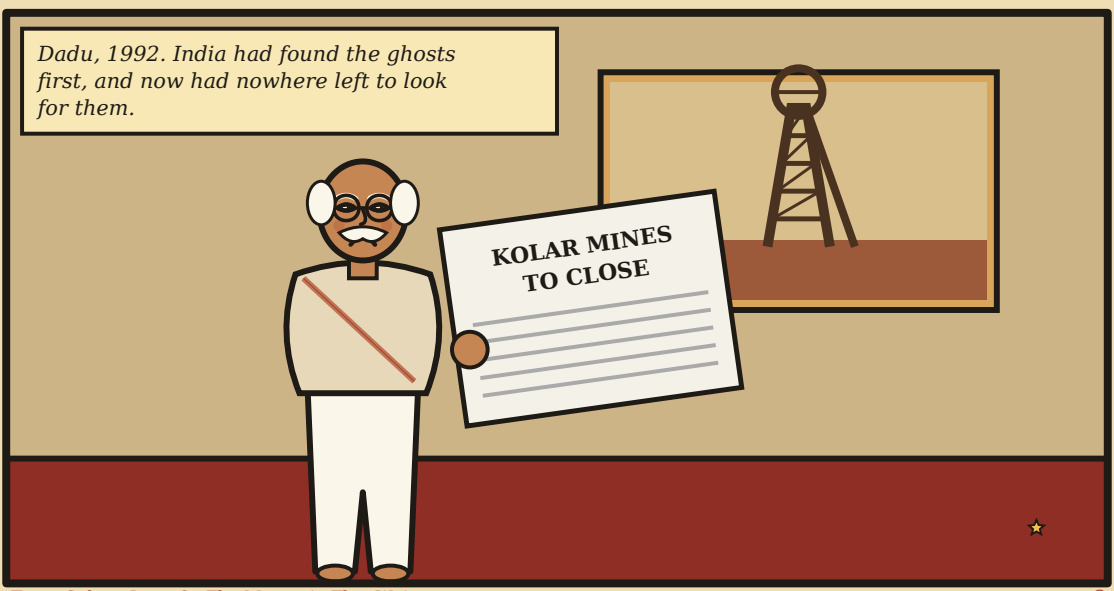
But the gold ran low. In 1992 the mines closed.



Water filled the tunnels. The lab was gone.

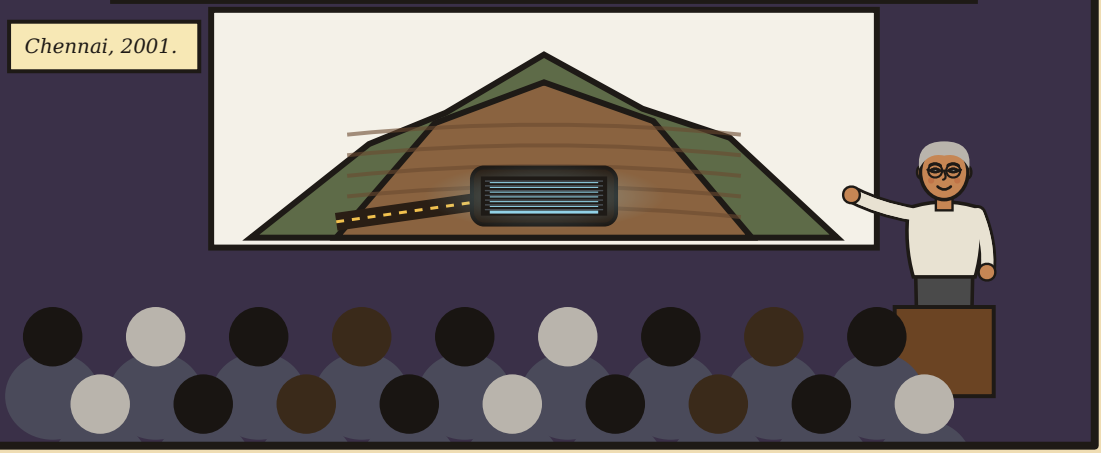


Dadu, 1992. India had found the ghosts first, and now had nowhere left to look for them.



NEUTRINO 2001 · CHENNAI

Chennai, 2001.





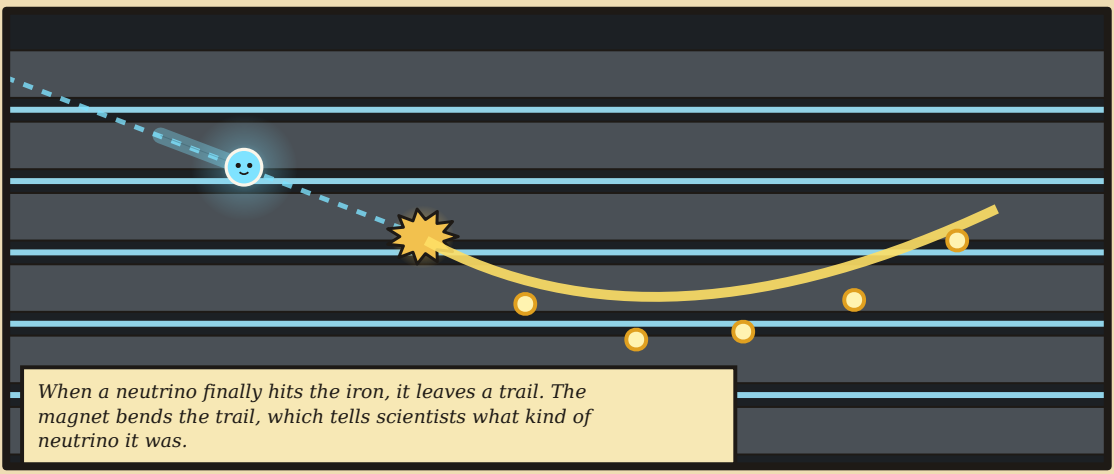
In the back row, a student named Meera.

Indian physicists decided to build again. Not in a mine this time, but inside a mountain. A giant detector called ICAL.

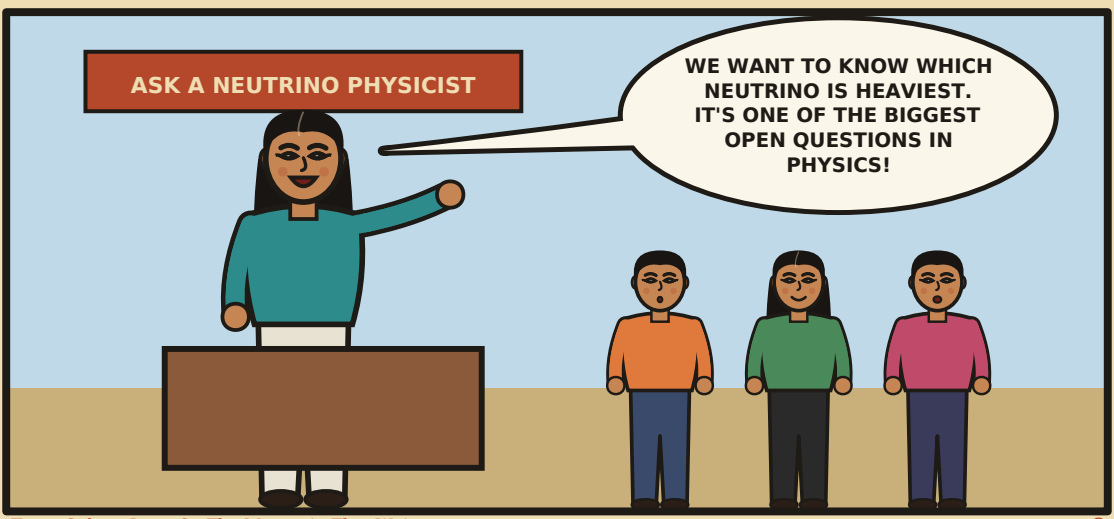


50,000 tonnes of magnetized iron. Heavier than six Eiffel Towers.

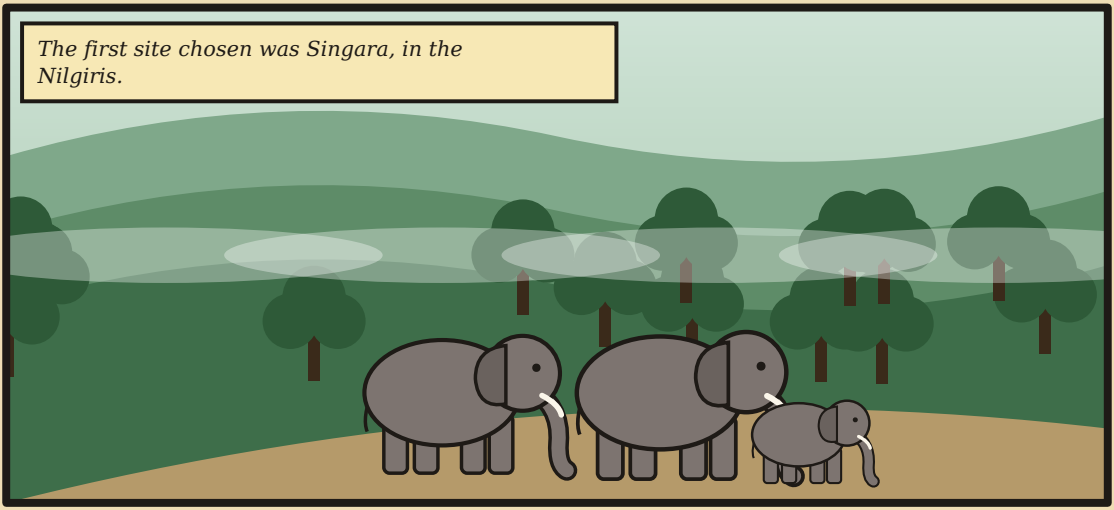


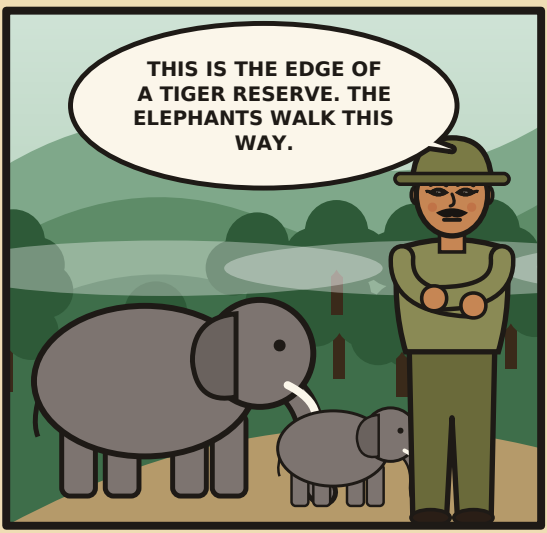


When a neutrino finally hits the iron, it leaves a trail. The magnet bends the trail, which tells scientists what kind of neutrino it was.



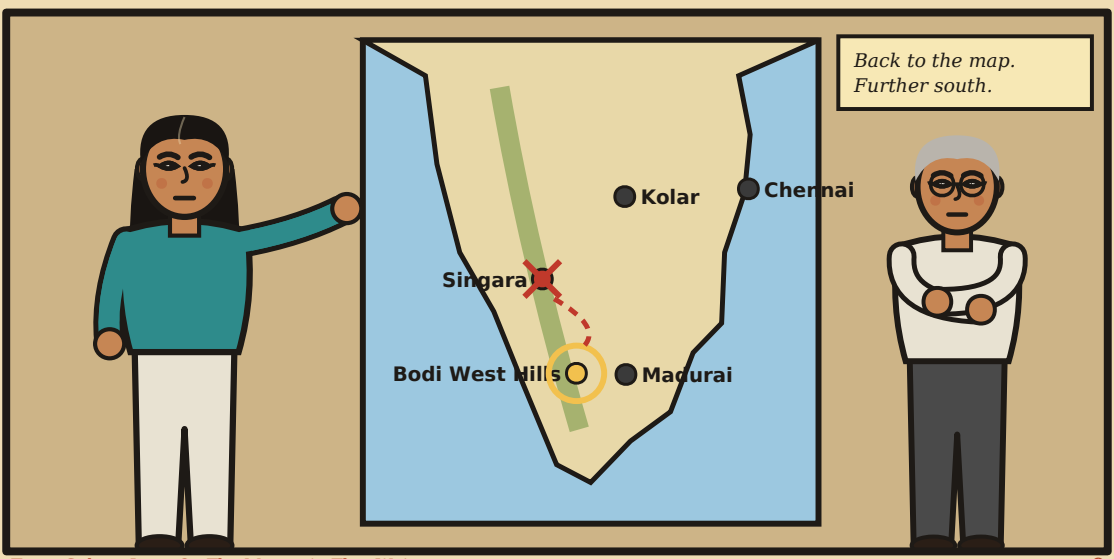
The first site chosen was Singara, in the Nilgiris.



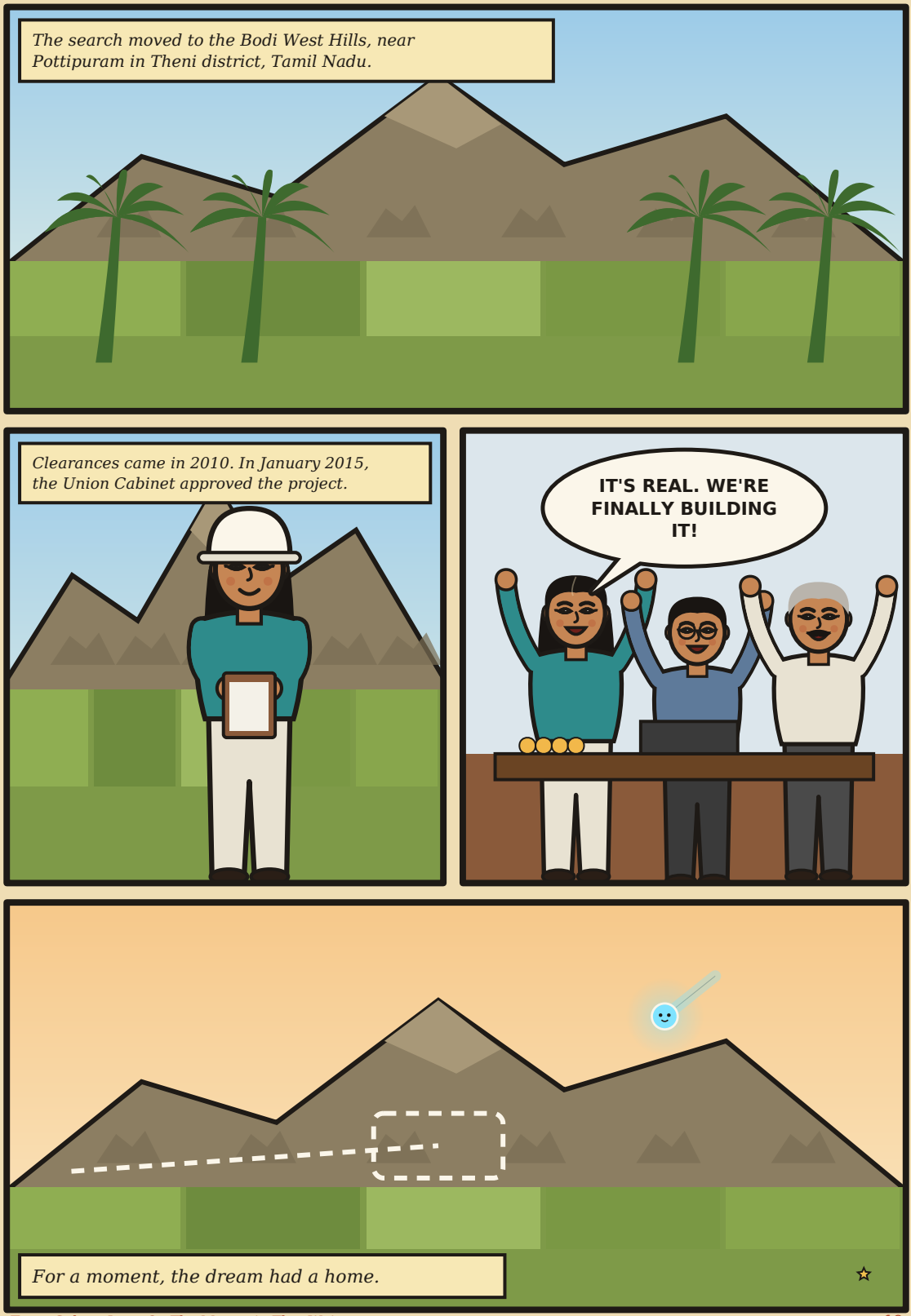




In 2009, permission was refused. The site lay in the buffer zone of the Mudumalai Tiger Reserve.



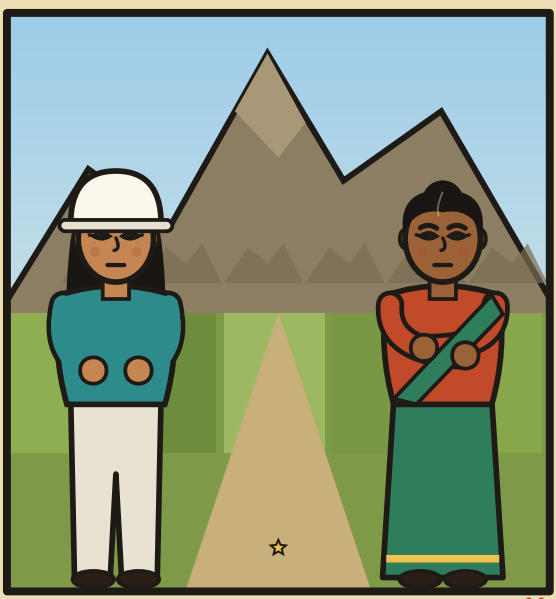
Back to the map. Further south.

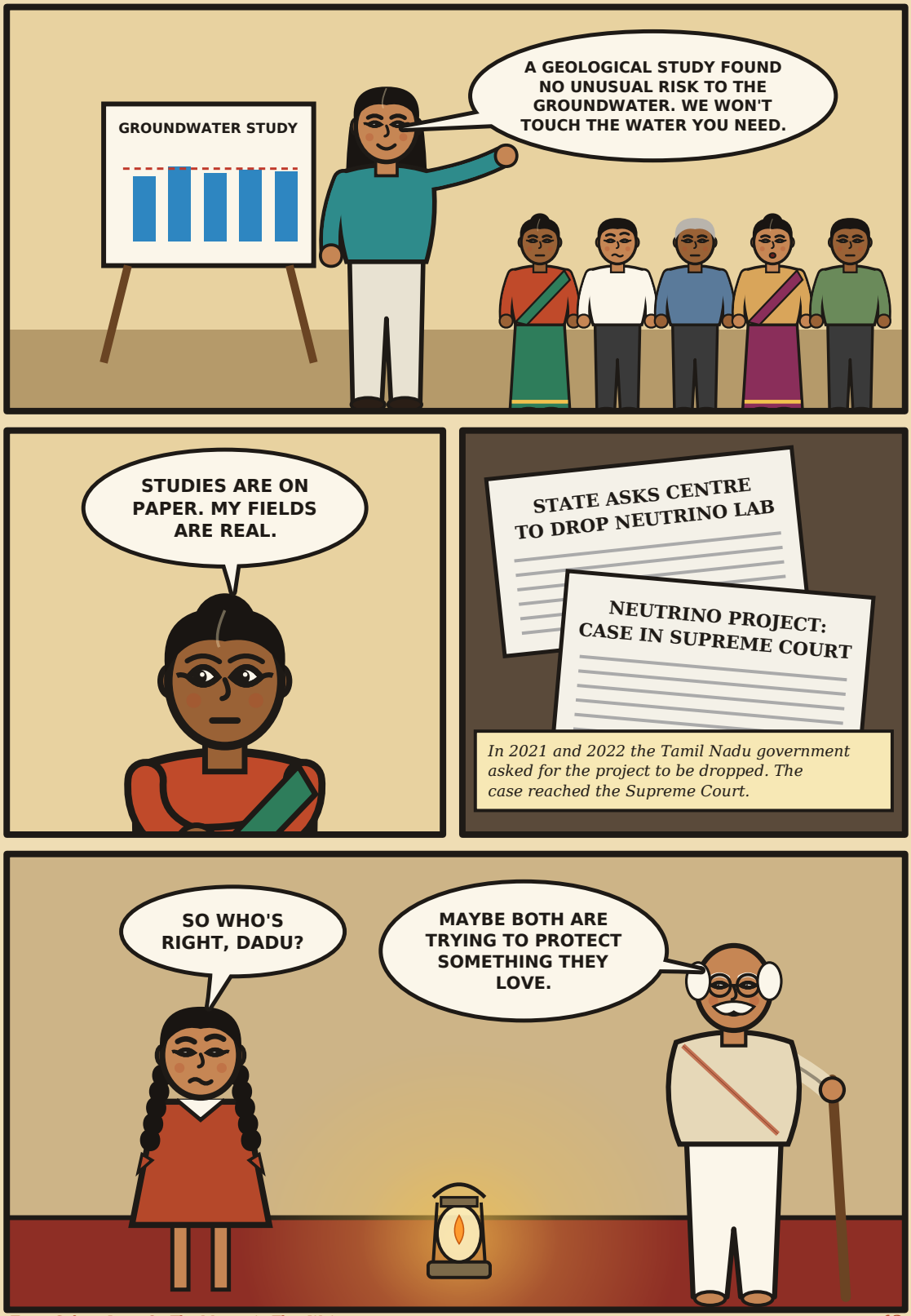

The search moved to the Bodi West Hills, near Pottipuram in Theni district, Tamil Nadu.
Clearances came in 2010. In January 2015, the Union Cabinet approved the project.
IT'S REAL. WE'RE FINALLY BUILDING IT!
For a moment, the dream had a home.


THEY'LL BLAST A TUNNEL INTO OUR MOUNTAIN. WHAT IF OUR WELLS DRY UP?


Local people and environmentalists worried about water, earthquakes, forests and the Western Ghats.


They went to court. In 2015, work was halted until more permissions were obtained.




GROUNDWATER STUDY
A GEOLOGICAL STUDY FOUND NO UNUSUAL RISK TO THE GROUNDWATER. WE WON'T TOUCH THE WATER YOU NEED.
STUDIES ARE ON PAPER. MY FIELDS ARE REAL.
STATE ASKS CENTRE TO DROP NEUTRINO LAB
NEUTRINO PROJECT: CASE IN SUPREME COURT
In 2021 and 2022 the Tamil Nadu government asked for the project to be dropped. The case reached the Supreme Court.
SO WHO'S RIGHT, DADU?
MAYBE BOTH ARE TRYING TO PROTECT SOMETHING THEY LOVE.

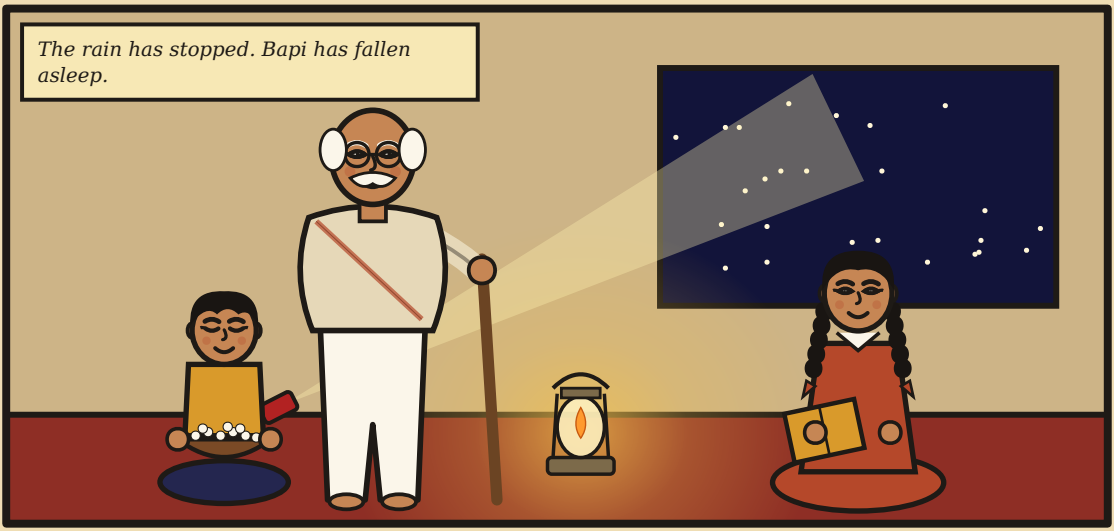
Madurai.

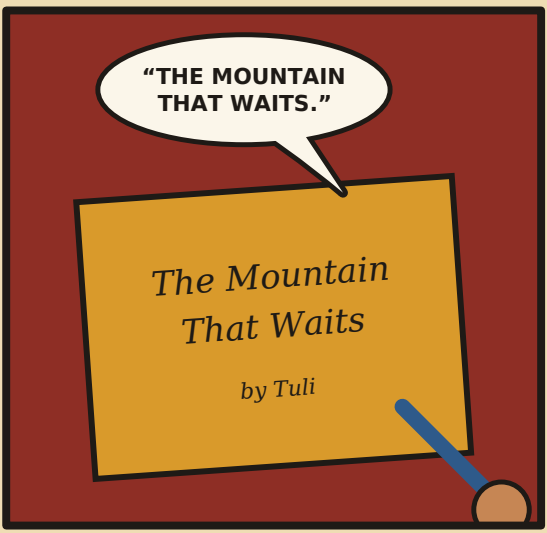
The mountain lab was never built. But a small prototype, mini-ICAL, has been running since 2018.

EVERY DAY IT TEACHES US HOW TO BUILD THE REAL ONE.

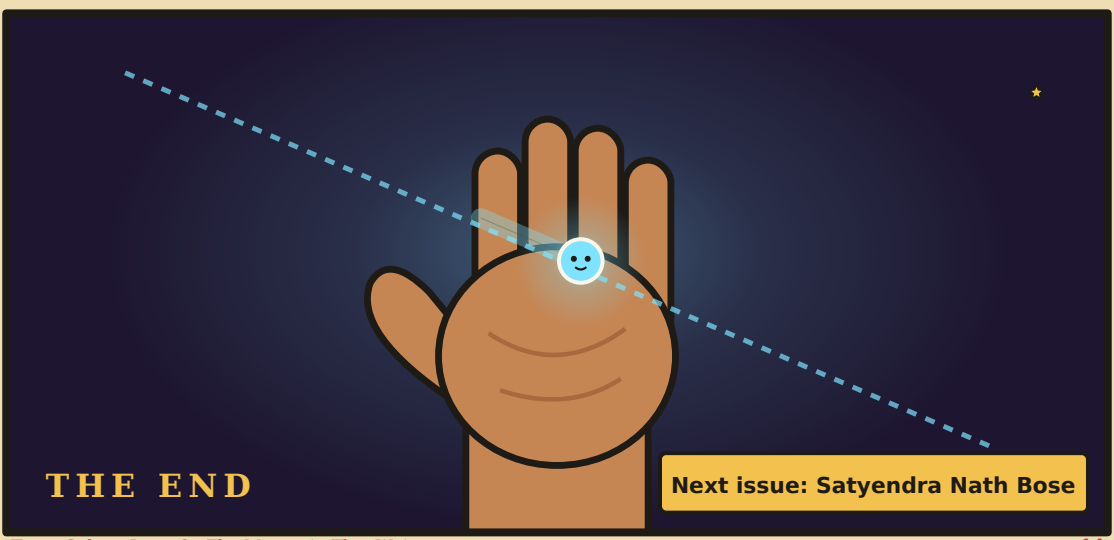
Young physicists trained here now work on neutrino experiments around the world.

The rain has stopped. Bapi has fallen asleep.









FACT FILE

What really happened

TIMELINE

1965 Atmospheric neutrinos detected deep in the Kolar Gold Fields by a TIFR–Osaka–Durham team.*

1989 First discussions about an Indian neutrino observatory.

1992 Kolar mines close; the underground lab ends.*

2001 INO collaboration formed at a neutrino meeting in Chennai.

2002 Feasibility-study agreement signed (30 Aug).

2009 Singara site refused: buffer zone of Mudumalai Tiger Reserve.

2010 Clearance for the Bodi West Hills site, Theni.

2015 Union Cabinet approves INO (Jan). Madras High Court halts work pending permissions (Mar).

2018 mini-ICAL prototype begins recording cosmic muons.

2021–22 Tamil Nadu asks for the project to be dropped; case before the Supreme Court.

WORDS TO KNOW

Neutrino
A tiny, nearly massless particle that almost never interacts with matter.

Atmospheric neutrinos
Neutrinos made when cosmic rays smash into the air high above us.

ICAL
INO's planned Iron Calorimeter: 50,000 tonnes of magnetized iron layered with glass detectors.

Mass ordering
Which neutrino is heaviest? One of the big open questions ICAL was designed to study.

BY THE NUMBERS

50,000 tonnes: planned weight of ICAL

~1.3 km: rock above the planned lab

~2 km: planned access tunnel

2018: mini-ICAL starts taking data

WHERE THINGS STAND

The latest status we found (2022): the case was before the Supreme Court and no construction had taken place at the site. Check the news for anything newer.

READ MORE

Wikipedia: “India-based Neutrino Observatory”

physicsworld.com: “India-based neutrino observatory faces a new hurdle”

downtoearth.org.in: “Tamil Nadu says no to INO in Theni”

india.mongabay.com (April 2022)

** Kolar details from general sources; check before citing.*

Meera, Selvi, Ranger Kumar, Miner Chacha, Dadu, Tuli and Bapi are fictional characters.

Story & concept: Haradhan Adhikary